

problem 1.

~~Assume~~

Let $G = e^{tX}$, $H = e^{tY}$

then:

$$\text{CDF: } F_G(g) = P(G \leq g) = P(e^{tX} \leq g).$$

$$F_H(h) = P(H \leq h) = P(e^{tY} \leq h)$$

① $t = 0$:

$G = e^{0 \cdot X} = 1$, $H = e^{0 \cdot Y} = e^0 = 1$, they are independent. ✓

② $t > 0$:

$$F_G(g) = P(e^{tX} \leq g) = P(tX \leq \ln g) = P(X \leq \frac{1}{t} \ln g)$$

$$F_H(h) = P(e^{tY} \leq h) = P(tY \leq \ln h) = P(Y \leq \frac{1}{t} \ln h)$$

$$F_{G,H}(g, h) = P(e^{tX} \leq g \cap e^{tY} \leq h) = P\left(X \leq \frac{1}{t} \ln g \cap Y \leq \frac{1}{t} \ln h\right)$$

Because X, Y are independent, then:

$$P\left(X \leq \frac{1}{t} \ln g \cap Y \leq \frac{1}{t} \ln h\right) = P\left(X \leq \frac{1}{t} \ln g\right) \cdot P\left(Y \leq \frac{1}{t} \ln h\right)$$

$$\Rightarrow F_{G,H}(g, h) = F_G(g) \cdot F_H(h)$$

$\Rightarrow G, H$ are independent. ✓

③ $t < 0$:

$$F_G(g) = P(e^{tX} \leq g) = P(X \geq \frac{1}{t} \ln g)$$

$$F_H(h) = P(e^{tY} \leq h) = P(Y \geq \frac{1}{t} \ln h)$$

$$F_{G,H}(g, h) = P\left(X \geq \frac{1}{t} \ln g, Y \geq \frac{1}{t} \ln h\right)$$

Because X, Y are independent, then:

$$F_{G,H}(g, h) = P\left(X \geq \frac{1}{t} \ln g\right) \cdot P\left(Y \geq \frac{1}{t} \ln h\right) = F_G(g) \cdot F_H(h)$$

$\therefore \forall$ constant $t \in \mathbb{R}$,

$$F_{G,H}(g, h) = F_G(g) \cdot F_H(h)$$

$\Rightarrow e^{tX}, e^{tY}$ are independent. ✓

Nice!

problem 2

Assume $X \sim N(\mu, \sigma^2)$, $Z \sim N(0, 1)$

then :

$$Z = \frac{X - \mu}{\sigma}$$

$$\Rightarrow X = \sigma \cdot Z + \mu$$

smart idea! ✓

$$E[Z^k] = \int_{-\infty}^{+\infty} z^k \cdot f_Z(z) dz$$

Because $f_Z(z)$ is an even function

so, $z^k \cdot f_Z(z)$ is a odd function, when k is odd number

and $E[Z^k] = 0$, $k = 2n+1$ ($n=0,1,\dots$) ✓

$$E[Z^2] = \int_{-\infty}^{+\infty} z^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{(z-\mu)^2}{2\sigma^2}} dz$$

$$= \int_{-\infty}^{+\infty} z^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$= z \cdot (-e^{-z^2/2}) \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} (-e^{-z^2/2}) dz$$

$$= 0 + 1$$

$$= 1$$

$$E[Z^4] = \int_{-\infty}^{+\infty} z^4 \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

$$= \int_{-\infty}^{+\infty} z^3 (-e^{-z^2/2}) \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} (-e^{-z^2/2}) (3z^2) dz$$

$$= 0 + 3 \int_{-\infty}^{+\infty} z^2 \cdot e^{-z^2/2} dz$$

$$= 3E[Z^2]$$

$$= 3$$

$$\therefore E[X] = \mu$$

$$E[X^2] = E[(\sigma Z + \mu)^2] = \mu^2 + 2\mu\sigma E[Z] + \sigma^2 E[Z^2] = \mu^2 + \sigma^2$$

$$E[X^3] = E[(\sigma Z + \mu)^3] = \mu^3 + 3\mu^2\sigma E[Z] + 3\mu\sigma^2 E[Z^2] + \sigma^3 E[Z^3] = \mu^3 + 3\mu\sigma^2$$

$$E[X^4] = E[(\sigma Z + \mu)^4] = \mu^4 + 4\mu^3\sigma E[Z] + 6\mu^2\sigma^2 E[Z^2] + 4\mu\sigma^3 E[Z^3] + \sigma^4 E[Z^4]$$

$$= \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$$

Excellent!

problem 3.

① X is a constant with probability 1 $\Rightarrow E[\phi(X)] = \phi(E[X])$

assume $X = c$, then:

$$E[\phi(X)] = E[\phi(c)] = \phi(c)$$

$$E[X] = E[c] = c \Rightarrow \phi(E[X]) = \phi(c)$$

$$\therefore E[\phi(X)] = \phi(E[X])$$

② $E[\phi(X)] = \phi(E[X]) \Rightarrow X$ is a constant with probability 1

$\because \phi$ is a strictly convex function

$\therefore$ ~~then~~ At point $x = \mu$:

$$\phi(X) - \phi'(\mu)(X - \mu) - \phi(\mu) = 0$$

when $X \neq \mu$:

$$\phi(X) - \phi'(\mu)(X - \mu) - \phi(\mu) > 0$$

$$\Rightarrow E[\phi(X) - \phi'(\mu)(X - \mu) - \phi(\mu)] > 0$$

$$\Rightarrow E[\phi(X)] - \phi'(\mu)E[X - \mu] - E[\phi(\mu)] > 0$$

$$\underline{E[X] = \mu}, E[\phi(X)] - 0 - E[\phi(\mu)] > 0$$

$$\Rightarrow E[\phi(X)] > E[\phi(\mu)] = \phi(\mu) = \phi(E[X])$$

$\therefore$ when $E[\phi(X)] = \phi(E[X])$, then:

$$X = \mu \Rightarrow \phi(X) - \phi'(\mu)(X - \mu) - \phi(\mu) = 0$$

only $X = \mu \Rightarrow Y = 0$, otherwise $Y > 0 \Rightarrow E[Y] > 0$

$$\text{only } X = \mu \Rightarrow Y = 0 \Rightarrow E[Y] = E[\phi(X) - \phi'(\mu)(X - \mu) - \phi(\mu)] = 0 \Rightarrow E[\phi(X)] = \phi(E[X])$$

otherwise $X \neq \mu \Rightarrow Y > 0 \Rightarrow E[Y] > 0$

$\therefore E[\phi(X)] = \phi(E[X]) \Rightarrow X$ is a constant with probability 1

Perfect!

problem 4.

~~Let~~

① Geometric distribution

Let $X_1, X_2, \dots, X_n$ be i.i.d., the PMF:

$$P(X=k) = (1-p)^{k-1} \cdot p, \quad k=1, 2, \dots$$

the likelihood function:

$$\begin{aligned} L(p | X_1, \dots, X_n) &= \prod_{i=1}^n P(X_i = x_i) \\ &= \prod_{i=1}^n (1-p)^{x_i-1} p \\ &= p^n (1-p)^{\sum_{i=1}^n (x_i-1)} \\ &= p^n (1-p)^{\sum_{i=1}^n x_i - n} \end{aligned}$$

$$\Rightarrow \ln L(p) = \ln L(p | X_1, \dots, X_n) = n \ln p + (\sum_{i=1}^n x_i - n) \ln(1-p)$$

* Take the derivative with respect to p and set it to 0.

$$\frac{dL}{dp} = \frac{n}{p} + (\sum_{i=1}^n x_i - n) \cdot (-\frac{1}{1-p}) = 0$$

$$\Rightarrow n(1-p) = p(\sum_{i=1}^n x_i - n)$$

$$\Rightarrow \hat{p} = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{x}}, \quad \bar{x} \text{ is the sample mean}$$

$$\because E[1/X] = 1/p, \text{ so } E[\bar{x}] = 1/p$$

assume $g(x) = 1/x$, this function is convex on the domain of $x > 0$.

(Because $g''(x) = 2/x^3 > 0$).

According to Jensen's inequality: $E[g(x)] \geq g(E[x])$.

Let $x = \bar{x}$, then:

$$E[\frac{1}{\bar{x}}] \geq 1/E[\bar{x}] \quad E[\frac{1}{\bar{x}}] \geq 1/E[\bar{x}]$$

$$\Rightarrow \hat{p} = E[\frac{1}{\bar{x}}] \geq 1/E[\bar{x}] = 1/(1/p) = p$$

$$\Rightarrow \hat{p} \geq p$$

so, $\hat{p}$ is biased

problem 4

③ Exponential distribution:

Let $x_1, x_2, \dots, x_n$ be i.i.d and $x_i \sim \text{Exp}(\lambda)$.

$$\text{PDF: } f(x; \lambda) = \lambda e^{-\lambda x} \quad x > 0$$

the likelihood function:

$$L(\lambda | x_1, \dots, x_n) = \prod_{i=1}^n f(x_i; \lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda(\sum x_i)}$$

$\Rightarrow$

$$\ell(\lambda) = \ln L(\lambda | x_1, \dots, x_n) = n \ln \lambda - \lambda \sum x_i$$

take the derivator with respect to λ and set it to 0

$$\frac{n}{\lambda} - \sum x_i = 0$$

$$\Rightarrow \hat{\lambda} = \frac{n}{\sum x_i} = \frac{1}{\bar{x}} \quad (\bar{x} \text{ is the sample mean})$$

Assume $g(x) = \frac{1}{x}$, $\therefore g'(x) = -\frac{1}{x^2} < 0$

$\therefore g(x)$ is a convex function in the domain of $x > 0$

According to Jensen's inequality: **Excellent!**

$$E[g(x)] \geq g(E(x))$$

Let $x = \bar{x}$, then:

$$E[g(\bar{x})] = E\left[\frac{1}{\bar{x}}\right]$$

$$g(E[\bar{x}]) = g(E[\bar{x}]) = \frac{1}{E[\bar{x}]}$$

$$\therefore E\left[\frac{1}{\bar{x}}\right] \geq \frac{1}{E[\bar{x}]}$$

$$\Rightarrow \hat{\lambda} = E\left[\frac{1}{\bar{x}}\right] \geq \frac{1}{E[\bar{x}]} = \frac{1}{1/\lambda} = \lambda$$

when $\bar{x}$ is a constant: $E[\bar{x}] = \bar{x}$

~~but~~ since $\bar{x}$ is not a constant, the inequality is strictly:

$$\hat{\lambda} > \lambda$$

$\therefore \hat{\lambda}$ is biased.

problem 5.

The likelihood function.

$$L(\lambda) = \prod_{i=1}^n P(X=x_i) = \prod_{i=1}^n \frac{e^{-\lambda} \cdot \lambda^{x_i}}{x_i!} = \frac{e^{-\lambda n} \cdot \lambda^{\sum x_i}}{\prod_{i=1}^n (x_i)!}$$

~~take derivator with respect to λ and set it to 0.~~

$$l(\lambda) = \ln L(\lambda) = -\lambda n + \sum x_i \ln \lambda - \ln \prod_{i=1}^n (x_i)!$$

take derivator with respect to λ and set it to 0.

$$-n + \frac{1}{\lambda} \sum x_i = 0$$

$$\Rightarrow \hat{\lambda} = \frac{1}{n} \sum x_i = \bar{x} \quad (\bar{x} \text{ is the sample } \text{mean} \text{ even}).$$

$$\psi = P(X=0) = e^{-\lambda}$$

substituting $\hat{\lambda} = \bar{x}$ into ψ :

$$\hat{\psi} = e^{-\hat{\lambda}} = e^{-\bar{x}}, \quad \hat{\psi} \text{ is a convex function.}$$

$E[\hat{\psi}] = E[e^{-\bar{x}}]$. According to Jensen's Inequality.

$$E[\hat{\psi}(\bar{x})] \geq \hat{\psi}(E[\bar{x}]) = e^{-E[\bar{x}]}$$

$$\Rightarrow E[e^{-\bar{x}}] \geq e^{-E[\bar{x}]}$$

$$\Rightarrow E[e^{-\bar{x}}] \geq e^{-\lambda}$$

$$\Rightarrow E[\hat{\psi}] \geq e^{-\lambda}$$

since e^{-x} is a strictly convex function. $\Rightarrow E[\hat{\psi}] \neq e^{-\lambda}$

$$\text{so, } E[\hat{\psi}] > e^{-\lambda}$$

$\therefore \hat{\psi} = e^{-\bar{x}}$ is ~~not~~ biased.

Nicely Done!

problem 6:

Assume $Y = X - \mu$

$\therefore X > \mu, \therefore Y > 0$

PDF: $f_Y(y) = f(X; \lambda, \mu) = \lambda e^{-\lambda y} \quad y > 0$

$\therefore Y \sim \text{EXP}(\lambda)$

$$\therefore E[Y] = \frac{1}{\lambda}, \quad E[Y^2] = \text{Var}(Y) + (E[Y])^2 = \frac{1}{\lambda^2} + \frac{1}{\lambda^2} = \frac{2}{\lambda^2}$$

$$\therefore E[X] = E[Y + \mu] = \frac{1}{\lambda} + \mu \quad (*)$$

$$\text{Var}(X) = \text{Var}(Y + \mu) = \text{Var}(Y) = \frac{1}{\lambda^2}$$

$$\therefore s^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2, \quad \text{Let } s^2 = \text{Var}(X):$$

$$\text{assuming } s^2 = \frac{1}{\lambda^2} \Rightarrow \tilde{\lambda} = \frac{1}{s}$$

substituting $\lambda = \frac{1}{s}$ into (*):

$$E[X] = \frac{1}{1/s} + \mu = s + \mu$$

$$\text{Let } E[X] = \frac{1}{n} \sum X_i = \bar{X}:$$

$$\therefore s + \mu = \bar{X} \Rightarrow \tilde{\mu} = \bar{X} - s$$

$$\therefore \tilde{\lambda} = \frac{1}{s}, \quad \tilde{\mu} = \bar{X} - s$$

Perfect!

② MLE:

$$L(\lambda, \mu) = \prod_{i=1}^n f(X_i; \lambda, \mu) = \prod_{i=1}^n \lambda e^{-\lambda(X_i - \mu)} = \lambda^n e^{-\lambda \sum (X_i - \mu)}$$

$$\ell(\lambda, \mu) = \ln L(\lambda, \mu) = n \ln \lambda - \lambda \sum_{i=1}^n (X_i - \mu) = n \ln \lambda + n \lambda \mu - \lambda \sum_{i=1}^n X_i$$

take derivator with respect to λ and set it 0

$$\frac{n}{\lambda} + n\mu - \sum_{i=1}^n X_i = 0$$

$$\Rightarrow \hat{\lambda} = 1 / \left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \right)$$

For a fixed $\lambda > 0$, $e^{-\lambda(X_i - \mu)}$ is an increasing function of μ . To maximize $L(\lambda, \mu)$, we need to make μ as large as possible. But $\mu \leq X_i \Rightarrow \mu = \min\{X_1, X_2, \dots, X_n\}$

$$\therefore \hat{\mu} = \min(X_1, X_2, \dots, X_n), \quad \hat{\lambda} = 1 / \left(\frac{1}{n} \sum_{i=1}^n X_i - \min(X_1, X_2, \dots, X_n) \right)$$

You have done outstanding work!
Keep it up!